Learning Closed Loop Kinematic Controllers for Continuum Manipulators in Unstructured Environments

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Abstract: This paper introduces a machine learning based approach for closed loop kinematic control of continuum manipulators in the task space. For this purpose we propose a unique formulation for learning the inverse kinematics of a continuum manipulator while integrating end effector feedback. We demonstrate that this model-free approach for kinematic control is very well suited for nonlinear stochastic continuum robots. Specifically, the paper addresses problems which are vital for practical realization of machine learning techniques. The primary objective is to solve the redundancy problem while making the algorithm scalable, fast, tolerant to stochasticity, requires minimal sensor elements and involves few open parameters for tuning. In addition, we present that the proposed controller can exhibit adaptive behaviour in the presence of external forces and unstructured environment with the help of the morphological properties of the manipulator. Experimental validation of the proposed controller is done on a 6 Degree of Freedom tendon driven manipulator control of the end effector in three dimensional space with and without external forces. The experimental results exhibit accurate, reliable and adaptive behaviour of the proposed system which appears suitable for the field of continuum service robots.

INTRODUCTION

The growth of soft robotics has been driven by new scientific paradigms and applications which offers a radical technological solution for well-known robotics problems. It has led to an alternate view on intelligence, incorporating the role of the physical body for fast, efficient and robust behaviour.[1,2] Recent years have seen the increasing trend of employing soft robots for grasping tasks[4,10], adaptive locomotion in land[5,6,7] and water[3,11], flexible prosthesis[8], and combined tasks.[9] Most of the researches in the field of soft robotics is currently being done on their design, material and fabrication. However, they are still lacking practical controllers that are employable in the real world.

Many of the traditional approaches for design, modelling and control, which have been so far effective for rigid robots, have become ineffective or incompetent for soft robots. The most commonly used kinematic modelling approach for soft robots is based on the piecewise constant curvature approximation.[12] More rigorous geometrically exact models were proposed by Renda et al.[13,14] However, unlike rigid robots, the kinematic models of soft robots are also dependent on environmental variables because of their numerous underactuated and compliant degrees of freedom (DoF). The existing controllers based on forward models are either too simple to account for external constraints or become too complex with the addition of external constraints.

There have been surprisingly few works on task space control of continuum manipulators in an unstructured environment even though they are potentially better suited than rigid robots in such conditions.[15] One of the first task space controllers with sensory feedback was proposed by Camarillo et al.[16] using analytical models to estimate the forward kinematics. Similarly, Kapadia et al. later introduced a model based task space controller assuming that the kinematic and dynamic model can be obtained and easily observable.[17,20] Joint space controllers based on the constant curvature approximation was proposed by Penning et al.[18,19] However, all these approaches still relied on analytical models which are heavily prone to modelling error and are invalid in the presence of environmental constraints and hence not suitable for more practical applications. Furthermore, system identification for the developed models is cumbersome and susceptible to errors.

Correspondingly, Marchese et al. developed closed loop control algorithms for fully soft manipulators based on piecewise constant curvature approximation[21] and extended it for whole arm motion planning in a known environment.[22] Interestingly, the motion planning was done considering the manipulator to adapt to the environment. Again, such methods are heavily dependent on the model formulation and sensing capabilities and based only on the closed loop

control of the segment curvature and not the end effector position. Alternatively, real-time control of soft robots were successfully implemented using a finite element based method (FEM) by Duriez.[23] Although they promise to be a very accurate representation of the actual model and can possibly incorporate known environmental constraints, their scalability to higher order systems is questionable. Furthermore, FEM based approaches become much more computationally expensive if they are to model external contacts too.

On the other hand, the underlying complexity and variability of continuum robots has prompted researchers to investigate the viability of model-free methodologies for control. If it is recognized that accurate analytical models are impossible to develop for any practical application, it is reasonable to lean towards model free methods for modelling and control. The first model-free controller for continuum robot was formulated by Yip et al. using empirical estimates of the Jacobian for task space control.[24,29] Although, these type of controllers are effective in unstructured environments, they require online estimation of the Jacobian and hence impractical.

Model-free controllers based on machine learning are a promising alternative considering their potential to adapt to altering conditions and generalize well between observed data even in the presence of noise. Not only is it applicable to a much broader variety of robots, they can also be faster to deploy and requires minimal sensory elements. The earliest usage of machine learning techniques in the field of continuum robots was used for the compensation of unknown dynamics of systems.[25] Later, Giorelli et al.[26] proposed the use of a feed forward neural network for learning the inverse statics of a soft cable driven three DoF manipulator. Their study indicated the effectiveness of machine learning based controllers over even a thoughtfully constructed analytical model for continuum robots. Even so, their approach cannot be applied to redundant or high dimensional continuum manipulators. Rolf et al.[27] presented a novel model called goal babbling for bootstrapping inverse models in high dimensional systems. Such an approach might scale well for high dimensional systems, although, not the most efficient. Yet, learning one solution at a time is restricting the full capability of a continuum robot and becomes invalid in the presence of external factors. Another interesting approach for learning the inverse kinematics was proposed by Melingui et al.[28] Their method involves learning the forward model and inverting it using distal supervised learning. Additionally, they have an adaptive subcontroller for compensating non-stationary kinetics. Again, these methods learn a particular solution to the IK problem and therefore become invalid in the presence of obstacles and moreover they do not scale well for higher dimensional systems.

Understandably, machine learning based approaches for obtaining inverse models are more developed for the conventional rigid robots. Thus, a small digression is made to comment on the most prevalent machine learning techniques developed for rigid robots. A common problem associated with learning the inverse kinematics (IK) of a redundant robot is the non-uniqueness of the solution. Coupled with the fact that the set of all the possible solutions form a concave set makes direct learning of the IK mapping hard.[30] Therefore it is necessary to have additional constraints/strategies to restrict/select the appropriate solutions. One of the earliest approaches for obtaining valid IK solutions by machine learning was proposed with distal supervised learning method.[31] A more recent and popular approach is to learn the inverse kinematics problem in the velocity level using differential inverse kinematics.[30,32] This is driven by the insight that the solution set of the differential IK problem form a convex set locally. However, working on a velocity level would require integration over a predefined path to get solutions to positional targets. Consequently, few researchers have tried to develop direct learning of the inverse kinematics solution on position level. Bocsi et al. suggested using structured output machine learning methods for selecting a particular solution from a set of all possible solutions.[33] Similarly, Vannucci et al. proposed a system using growing neural gas for learning the global inverse kinematics on position level.[34,35] Practical implementations of these methods have been limited because of their complex formulations. Although a direct translation of these approaches can be realized for continuum robots, it must be realized that the kinematic formulation of continuum robots is fundamentally different from rigid robots. Unlike rigid robots, the kinematic model of continuum robots is dependent on environmental factors also. None the less, the underlying scheme behind these methods can still be replicated and revised for our purposed.

This paper presents a machine learning based approach for obtaining the global inverse kinematics solution for a redundant continuum robot based on the author's previous work.[36] Developing on the previous work, an effective and unique technique for integrating feedback information about the tracking

error in the learned system is proposed. The proposed method promises a simple, fast and efficient learning of the global Inverse Kinematics, even in the presence of singularities without requiring any prior information about the robot and few open parameters for the user to tune. In addition, the learning is done between the task space and actuator space, therefore requiring minimal sensor components. The generalizing ability of the regression algorithm (artificial neural networks, in our case) along with an optimal IK formulation and sampling strategy leads to very less sample data requirement. Finally, we propose a unique formulation of the IK learning problem, which allows the derived task space kinematic controller to display intelligent behaviour in an unstructured environment. Experimental results on a 6 DoF continuum manipulator indicates that the proposed approach is promising for continuum robot control and exhibits interesting behaviour under external disturbances

The next section describes the structure of learning algorithm for the IK solver and the underlying formulas which makes the data driven learning feasible. Further, experiments on a simulated 9 DoF continuum manipulator is conducted to give empirical evidence of the efficacy of the feedback scheme and to showcase its effectiveness. A brief outlook into the experimental setup developed for the 6 DoF manipulator is then described followed by experimental characterization of the manipulator kinematics. Afterwards, the training method and performance is described. Description of the experiments and their corresponding results are then mentioned in the following section. Finally, a summary of the paper noting the advantages and disadvantages of the proposed method along with scope for improvements and possible developments in the future are stated in the last section.

FORMULATION OF THE LEARNING ARCHITECTURE

The objective of this paper is to develop kinematic controllers by developing models for the IK solutions of the continuum robot. The forward kinematics can be represented as a mapping between the actuator space (encoder value/length of cables, cable tension, pneumatic pressure, etc.), $\boldsymbol{q}$, to the end effector coordinates, $\boldsymbol{x}$. Assuming that there are no environmental constraints, the forward kinematics can be represented by:

$$\boldsymbol{x} = f(\boldsymbol{q}) \tag{1}$$

Where, $\boldsymbol{x} \in \Re^m$ is the position and orientation vector; $\boldsymbol{q} \in \Re^n$ is the vector containing the actuator variables; and f is some surjective function. The inverse kinematic model is a mapping between the end effector coordinates to the actuator variables. Direct inversion of the forward function is not possible when $m < n$, i.e. when the manipulator is redundant and the solution set of all possible solutions do not form a convex set. To simplify the inversion problem, the forward kinematics can be linearized at a point ($\boldsymbol{q^o}$), thereby obtaining the formulation;

$$\dot{\boldsymbol{x}} = J(\boldsymbol{q^o})\, \dot{\boldsymbol{q}} \tag{2}$$

Here, J is the Jacobian matrix that transforms actuator velocities, $\dot{\boldsymbol{q}}$ to end effector velocities, $\dot{\boldsymbol{x}}$. As was shown by D'Souza et al.[30], by spatially localizing the actuator variable $\boldsymbol{q}$, we can ensure convexity of the different IK problem and thereby making the learning problem tractable. By sacrificing a little on the accuracy, the differential kinematics can be approximated as;

$$\Delta \boldsymbol{x} \approx J(\boldsymbol{q^o})\Delta \boldsymbol{q} \tag{3}$$

$$J(\boldsymbol{q_i})\boldsymbol{q}_{i+1} \approx \boldsymbol{x}_{i+1} - f(\boldsymbol{q_i}) + J(\boldsymbol{q_i})\boldsymbol{q_i} \tag{4}$$

Where, $\boldsymbol{q}_{i+1}$ is the actuator configuration that archives the end effector position $\boldsymbol{x}_{i+1}$, while $\boldsymbol{q_i}$ is the current actuator configuration. This not only allows us to learn the Inverse Kinematics on a position level but also facilitates spatially localizing $\boldsymbol{q}$ by the sampling method rather than the learning architecture. The spatial localization can be done by ensuring that $| \boldsymbol{q}_{i+1} - \boldsymbol{q_i} |$ is bounded. Note that this will indirectly bound the end effector motion spatially too. From empirical data, it is recommended that: $| \boldsymbol{q}_{i+1} - \boldsymbol{q_i} | < \epsilon$, where ϵ is between 3-10% of the actuator range. For faster exploration higher ϵ is better, however a lower value provides better accuracy. Much lower values of ϵ can theoretically provide better accuracy, but, in reality environmental noise will overpower the information present in the data. The exploration strategy for collecting sample data is done by continuous motor babbling whilst ensuring the spatial locality of the actuator variable. Now we can employ any universal approximator to learn directly the mapping $(\boldsymbol{x}_{i+1}, \boldsymbol{q_i}) \rightarrow (\boldsymbol{q}_{i+1})$. In our case we are using a single hidden-layer artificial neural network

for this purpose. We are using tan-sigmoid transfer function in the hidden layer and a linear transfer function in the output layer. Since the data is naturally bounded by the actuator range and due to the tan-sigmoid transfer function, the learned network will always output a valid joint configuration thus ensuring valid outputs. Now the important concern is what the response of the learned network would be when the target positions $(\boldsymbol{x}_{i+1})$ cannot be achieved from the current joint configuration by a local motion (Since, the network is trained with data that are obtained by local motions only). One can expect the learned network to be similar to the form given below:

$$\boldsymbol{q}_{i+1} = G(\boldsymbol{x}_{i+1} - f(\boldsymbol{q}_i) + J\boldsymbol{q}_i) + (I_n - GJ)\boldsymbol{z} \quad (5)$$

Where,G, is a generalized inverse of $J(\boldsymbol{q}_i)$ and I_n is the identity matrix and $\boldsymbol{z}$ is an arbitrary n-dimensional vector. As noted in the authors' previous work[36], when the target inputted is not a local point, the network response just gets scaled as one would expect from equation (5), thereby bringing the end effector position closer to the target. Repeating the algorithm eventually leads to the network converging near the target position.

Since neural networks have the ability to generalize well, global learning of the IK can be done without exploring the complete actuator space, which is very large even for our 6 DoF manipulator. This is also aided by the fact that there is always high correlation between local Jacobians. But, it is important that the exploration process obtains data from the boundaries of task space for maximal utilization of the workspace.

As stated before, eventually we intend to use these manipulators in complex environments and it's necessary for our controllers to exploit the adaptability of the physical body. Simple controllers which learn the mapping $(\boldsymbol{x}_{i+1}, \boldsymbol{q}_i) \rightarrow (\boldsymbol{q}_{i+1})$ will fail in the presence of environmental constraints, because for continuum robots, the forward kinematic model is also dependent on the environment. This is due to the numerous underactuated and compliant joints present in a continuum robot. It is a hard task to model all the contacts and get the subsequent kinematic model. Therefore, we propose a simpler way to incorporate feedback correction of errors occurring due to unstructured environmental factors and propose that this simple strategy can perform well even without any modification of the learned network.

Consider a controller which learns the mapping regulated by a different formulation of equation (6) shown below:

$$\boldsymbol{q}_{i+1} = G(\boldsymbol{x}_{i+1} - \boldsymbol{x}_i) + \boldsymbol{q}_i + (I_n - GJ)\boldsymbol{z} \quad (6)$$

We can expect a network which learns the mapping $(\boldsymbol{x}_{i+1}, \boldsymbol{q}_i, \boldsymbol{x}_i) \rightarrow (\boldsymbol{q}_{i+1})$ to always move along the direction of the Jacobian scaled by the error in tracking $(\boldsymbol{x}_{i+1} - \boldsymbol{x}_i)$. For the case of rigid robots, the information obtained from the end effector position $\boldsymbol{x}_i$ is redundant, however, that is not the case for continuum robots. As the estimate of the Jacobian by the learned network is based only on the current joint configuration it will not be same as the actual Jacobian. However, we argue that even with this inaccurate estimate of the Jacobian is enough to force the motion of the end effector in following a path of minimum possible error under external constraints. In other words, we claim that in the presence of obstacles a decent strategy is to try to reach the target towards the current estimate of the Jacobian and the high dimensionality and compliance of the body will guide the manipulator in the best path as long as the controller tries to reduce the tracking error with the current estimate of the Jacobian. In addition, adding the end effector position as an input makes the learning more tolerant to noise incurred due to the stochasticity of the system, which greatly improves the learning process. Since, it is hard to predict the underlying algorithm embedded in the network after learning, testing and validation of the proposed learning scheme under extreme situations is done with the help of simulations in the next section

SIMULATIONS

In this section two simulations are performed to test and validate the scheme for integrating the end effector feedback to the learning scheme. Through these simulations we try to demonstrate that the proposed controller can accommodate changes in kinematics of the robot without retraining the network. The simulations are conducted on a kinematic model of the bionic handling assistant.[38] The continuum kinematics is modelled by a constant curvature approximation and the manipulator has nine DoFs.

To begin with data samples are generated by continuous motor babbling as mentioned in the previous section and the mapping $(\boldsymbol{x}_{i+1}, \boldsymbol{q}_i, \boldsymbol{x}_i) \rightarrow (\boldsymbol{q}_{i+1})$ is learned. During the learning phase no external disturbances are applied to the system and therefore information about the current end effector position $(\boldsymbol{x}_i)$ is actually redundant. None the less, our interest lies in studying the effects of external factors on the manipulator and the subsequent response of the proposed learning system.

Since the simulation model is purely kinematic, it is not possible to apply external forces directly. Therefore the only way to model external factors is to modify the forward kinematics of the model itself. Firstly, we set an offset to the end effector position outputted by the simulator and observe the performance of the solver. This is equivalent to adding a constant value to the forward kinematic model. Note that the offset is adding after learning and the IK solver does not have any information about the offset other than from the end effector position. The target position is fixed in all cases. Therefore, the solver has to output different actuator configurations for reaching the same target because of the different offsets. Figure 1 shows the error in tracking by the solver with the change in offset value added in the X direction. ϵ value is shown in percentage of the actuator range to showcase its effect on efficient learning. The number of samples collected is the same for all experiments. We can observe that for optimum value of ϵ ,the error in tracking can be reduced by the proposed feedback scheme. If there were to be no end effector feedback in the IK solver, the tracking error will be directly proportional to the offset distance as the end effector will always go the same actuator configuration, since the inputs remain the same.

Furthermore, we can realize more nonlinear variations in the forward model by modifying the cable lengths after learning the IK model. Figure 2 shows the performance of the IK solver with end effector feedback and the IK solver without feedback when the initial length of one of the nine cables is reduced. For more clarity, if the current cable length is q^o , the forward model always calculates the end effector position with the new cable length $q^o - K$ (K is a constant and denoted with respect to the initial cable length in Figure 2), while for the IK solver, the input still remains as q^o. For the case of the IK solver with end effector feedback, the position of the end effector along with the information from the initial kinematic model is enough for accurate tracking of the target. It must be brought to the attention of the reader that network is never explicitly directed to reduce the tracking error; the underlying error correction system is arising purely from the data and the learning formulation.

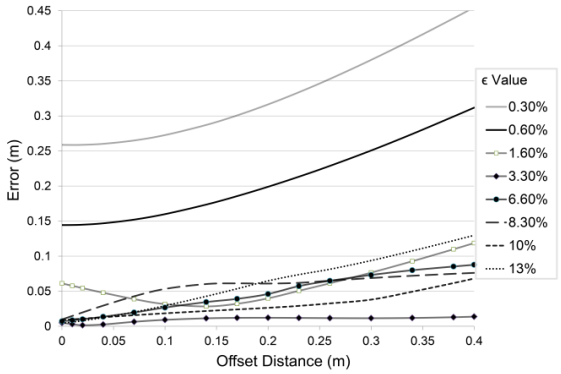


FIG. 1. Performance of the closed loop kinematic controller with offset added to kinematic model for different values of ϵ. The target is a fixed point for all cases.

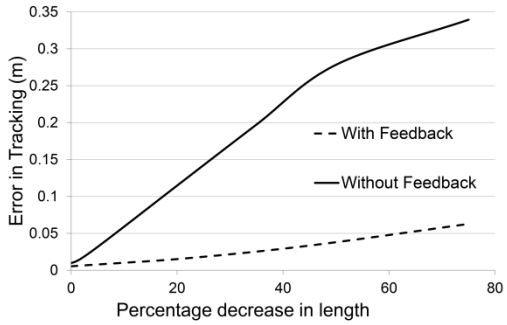


FIG. 2. Comparison of the performance of the closed loop kinematic controller and the open loop kinematic controller with non-linear changes in the forward kinematics. The target is a fixed point for all cases.

EXPERIMENTAL SETUP

For testing and validating the proposed learning controllers we use a recently developed modular continuum manipulator designed for showering application.[37] Since, the manipulator is developed for service applications, it is very important that the manipulator is inherently safe while the controller is reasonably accurate. The setup is composed of two hybrid modules with each modules having three pneumatic and three tendon-driven actuators. The McKibben-based flexible fluidic actuators are to be used in tandem with the inelastic cable driven actuators for extension, compression and stiffening. The actuators are supported externally by a flexible helicoidal structure that has been inserted along the entire module thus providing appropriate structural rigidity for our application while maintaining the dexterity of the arm. For our experiments we will be

using only cable driven actuation. Hence the manipulator has only six active degrees of freedom. The unactuated length of the manipulator is 40.5 *cm* with a diameter of 6 *cm*.

The cables are actuated by six HS-785HB Winch Servo Motors. Servo motors are used for their ease of control however they do cause undesirable jerky motion. For tracking the position and orientation of the manipulator the Aurora® tracking system (Northern Digital Inc.) is used with a six DoF electromagnetic probe. The probe is attached at the end of the manipulator. If the environment is free of electromagnetic disturbances, the system specifies an accuracy of 0.70 mm and 0.30 degrees (RMS). The Aurora system also specifies the uncertainty of each measurement which is useful during the learning step for removing outliers in the data. The manipulator setup for the experiments is shown in Figure 3.

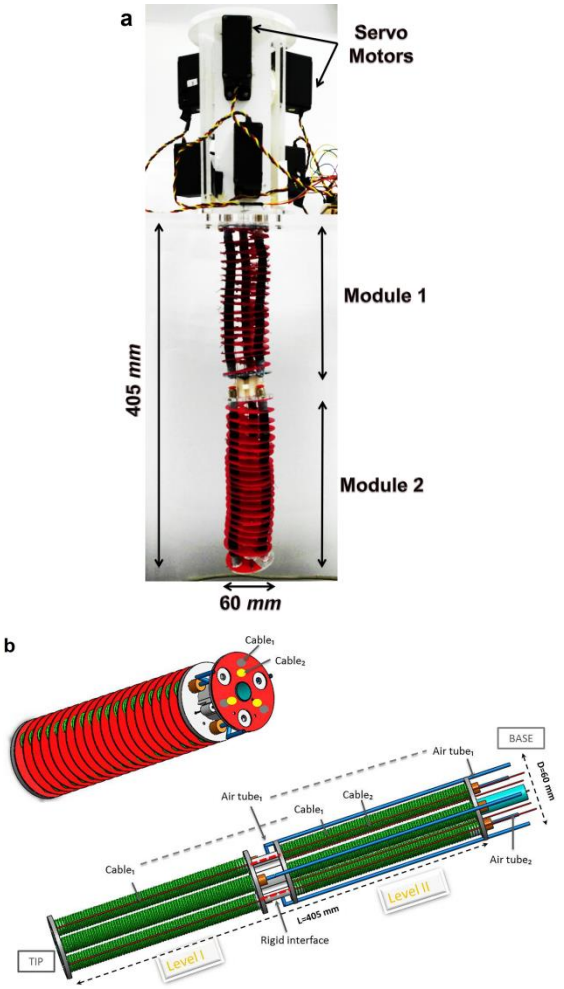


FIG. 3. **(a)** The 6 DoF continuum manipulator. **(b)** CAD model of the design.

MANIPULATOR CHARACTERIZATION

In this section a brief analysis of the manipulator characteristics is summarized. A representation of the manipulator workspace is shown in Figure 4 which is obtained through motor babbling. The un-actuated home position is marked with a larger circle. 2000 sample points were measured for this purpose. The same data will be used for the learning process, which is described in the next section. From direct observation of the workspace we can notice that the workspace is skewed. This is because of the asymmetric geometry of the manipulator, which arises from many factors. This is further indication about the importance of model-free learning in these scenarios. Also, few outliers can be observed near the boundaries as the probe also reaches the boundary of the electromagnetic field. Therefore during learning, each observation is prioritised with respect to the uncertainty in tracking, which is provided by the Aurora system. This will help reduce the effects of noise in tracking system. The main drawback of the manipulator design currently is the slacking of the cables due to the parallel configuration of the cables along the modules. This not only leads to kinematic singularities which reduce the performance of the IK solver but also condenses the reachable workspace of the manipulator. The singular configurations do not interrupt the learning process as they can be considered as the non-homogenous solutions to the non-homogenous equation 6.

Additionally, the stochasticity of the forward kinematic model is analysed to provide an indication of the best achievable accuracy of the inverse model without any feedback. This is done by first recording the coordinates of the end effector for 100 random joint configurations; repeating this step four times to get the variability in the data; calculating the mean value of the end effector position for each 100 joint configurations and calculating the residuals of each observations with respect to its corresponding mean. The standard deviation for all 400 residuals along with the mean of the absolute residuals is shown in Table 1. This can be considered as the expected performance of the IK solver without any feedback correction, provided that the learning is perfect with infinite sample data. Similar trends can be observed in the experimental results too, but with the aid of the feedback controller.

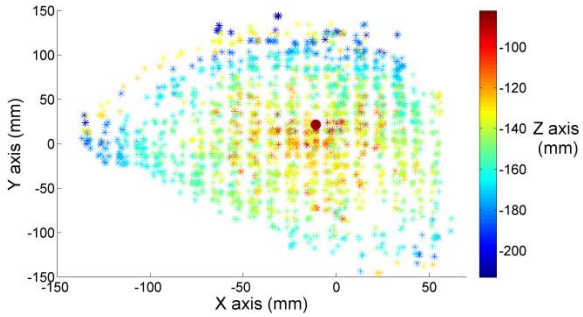


FIG. 4. Three dimensional workspace of the manipulator.

Table 1: Residuals of the repeatability test.

	Standard Deviation	Mean and Standard Deviation of Absolute Residuals
X (mm)	5.2	4.4±2.9
Y (mm)	3.9	3.1±2.3
Z (mm)	2.5	1.9±1.7
Yaw (degrees)	1.2	0.8±0.9
Pitch (degrees)	2.4	1.9±1.5
Roll (degrees)	2.8	2.3±1.7

LEARNING OVERVIEW

A multilayer perceptron with a single hidden layer is used for learning the mapping: $(\boldsymbol{x}_{i+1}, \boldsymbol{q}_i, \boldsymbol{x}_i) \rightarrow (\boldsymbol{q}_{i+1})$. Tan-sigmoid transfer function is employed in the hidden layer and a linear transfer function is employed in the output layer. For the implementation on the real manipulator we learn two IK solvers; one only for position and one for both position and orientation (Euler angles). Therefore we have one network with 12 inputs and another with 18 inputs. Both networks output the 6 motor positions. Bayesian regularization backpropagation algorithm is used for training the network. 2000 samples are collected for training and divided in the ratio 70:30 for training and testing. The ϵ value used for the exploration is 5% of the total actuator range. Each sample is collected after a delay of 2 seconds to ensure that manipulator comes to a complete rest. The uncertainty in each observation is used as a scaling factor in calculating the mean squared errors for reducing the effect of noise in the training process. Table 2 shows the training performance for each IK solver. In the whole learning process mentioned, there are only two open parameters for the user to decide. One is the exploration constant,ϵ, which is decided by the accuracy requirements and the stochasticity of the manipulator. Generally a value between 3-10% of the actuator range should guarantee valid learning. Additionally it is always possible to generate sample points for higher ϵ using data points of lower ϵ by sacrificing on the number of data points. The second open parameter is size of the neural network, which is easier to determine.

Table 2: Training Performance.

	IK solver for Position	IK solver for Pose
Layer Size	30	30
Epochs	61	138
Training NMSE	0.0086	0.0074
Testing NMSE	0.0099	0.0092
Training Time (s)	31	100

EXPERIMENTS AND RESULTS

In order to test and validate the performance of the solver five sets of experiments are conducted for real-time kinematic control of the manipulator with the two learned IK solver. The experiments are conducted for evaluating the performance of the solver for global point to point motion, local path following, local path following in an unstructured environment and disturbance rejection while maintaining a particular position. For all the experiments the manipulator starts from the home position, which is the configuration with all the actuators at the zero position. Figure 5 shows an overview of the IK solver based control system. The weights of the neural network are also not updated during any of the experiments.

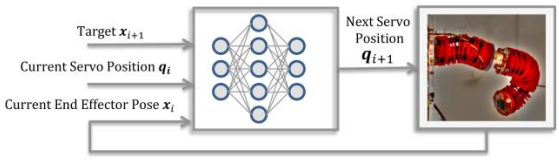


FIG. 5. Overview of the closed loop kinematic controller.

Point to Point motion for pose control

For the first experiment, twenty-five random points are selected from the manipulator workspace and the manipulator is commanded to reach each pose in succession using the solution provided by the IK solver. The points are selected from the workspace to ensure that the target is reachable as it is difficult to know beforehand whether a particular position and orientation is reachable. Since the targets are not local points, the solver will require few steps to converge. This makes the solutions fundamentally different from the training outputs and dependent on the initial conditions. Therefore it is possible to

obtain multiple solutions for the same target in case the manipulator is redundant for that case. The IK solver converges near the target within ten steps for all the targets. There is a delay of 1*sec* between each step during execution to ensure that the static condition is reached. Table 3 shows the average and standard deviations of the absolute errors incurred while tracking the targets.

Table 3: Point to Point Motion Performance

	Mean Error	Standard Deviation
Position (mm)	9.67	5.33
X (mm)	5.53	4.08
Y (mm)	5.04	4.59
Z (mm)	4.03	3.56
Yaw (degrees)	2.76	5.42
Pitch (degrees)	1.84	1.83
Roll (degrees)	3.85	7.02

Trajectory following with IK solver for position and pose

The next experiment is to test the performance of the solver in following a continuous path in the task space. Although the targets are local points, the difficulty lies in the fact that the corresponding actuator configurations need not be continuous in the actuator space. There are two trials conducted for this experiment. The first trial is conducted with the IK solver for position with the target path being a straight line of length 20 centimetres divided into 40 waypoints. The IK solver is given only one step to provide the solution for each waypoint with each step given 1 *sec* to settle. The second trial is conducted with the IK solver for pose with the same target positions and a fixed orientation of [90, 0, 180]. Figure 6 shows the errors in position and Figure 7,8 shows the errors in orientation of the manipulator in following the trajectory. The tracking error shown in Figure 7 is just for comparison with the controller for pose. The average positional error for line following using the kinematic controller for position is 23*mm* and the same for the pose controller is 37*mm*. Since the target is updated every step, the IK solver would not be able to converge to the best solution for each target and consequently the errors are higher for this experiment. In case of trajectory following with a fixed orientation it must be mentioned that some of the targets are statically unreachable. Our interest lies in analysing the IK solver even if the targets are not reachable.

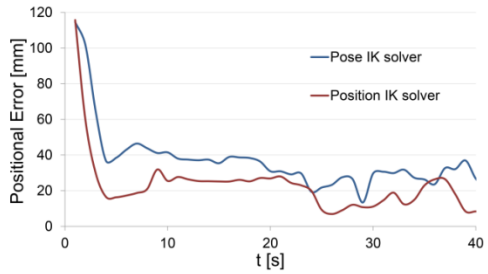


FIG. 6. Positional error in tracking for the two kinematic controllers.

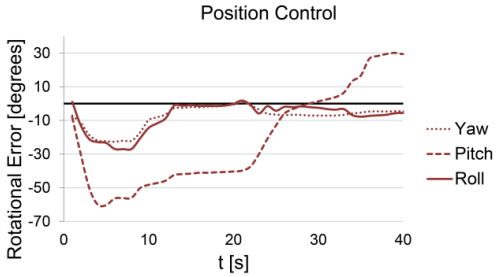


FIG. 7. Orientation error for kinematic controller for position given as a comparison for evaluating the kinematic controller for pose.

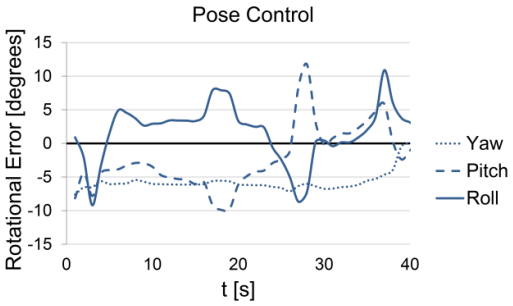


FIG. 8. Orientation error in tracking using the kinematic controller for pose.

Since the IK model is constructed by a neural network, the behaviour of the solver when the target is an outliers (For the case when the target is not reachable) is difficult to be foreseen. The underlying algorithm is majorly dependent on the training data. Figure 9 (a) shows the configurations of the manipulator during the path following task using the IK solver for position. It is clear that for the IK solver for pose it is impossible to reach the desired target during the beginning of the task and towards the end. The higher errors in position as shown in Figure 6 can be attributed to this. We can see that there is a trade-off between tracking the position and orientation of the robot. However, presently, it is difficult to quantify how the solver prioritizes for reducing the orientation and position tracking errors or even if there is such a trade-off internally.

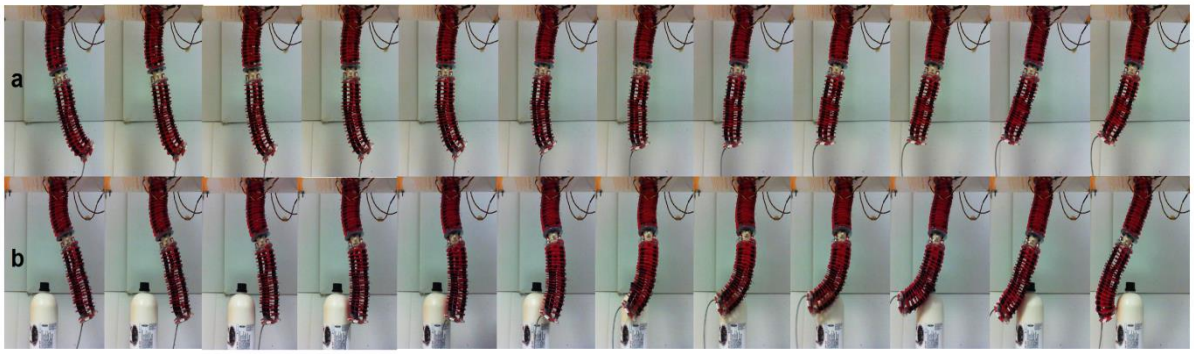


FIG. 9. (a) Configuration of the real manipulator at different time steps during the line following task.**(b)** Configuration of the real manipulator at the same corresponding time steps during the line following task in the presence of an obstacle.

Trajectory following in an unstructured environment

The next test is to see the effect of an unstructured environment while executing a task. The trajectory is the same straight line position path mentioned in the previous section but with just 20 points instead of 40. Each step is updated at 0.25 *sec*. This is to ensure that some of the momentum is still conserved between each step. A cylindrical obstacle of diameter 8*cm* is placed in between the path such that the centre of the cylinder lies around [10 -30] (Figure 10). We are only analysing the performance of the IK solver for position as the manipulator is only redundant for this task. Figure 9 (b) shows the configurations of the manipulator while performing the task. The numerous DoFs and compliance in the body aid the controller in tracking the best possible path under the circumstances. By a quick comparison with task shown in Figure 9(a) we can see the various kinematic configurations that the manipulator is able to achieve with the help of the environment. This is one of the key characteristics of continuum robots. The ability of continuum robots to have modified kinematics in the presence of external forces makes it almost impossible to get accurate formulation of the kinematics, on the contrary, it opens up the possibility of using the environment to guide the manipulator and facilitate the control objective. Through this experiment we demonstrate that approximate IK models with end-effector feedback can work in collaboration with the morphological properties of the robot to accomplish tasks even in an unstructured environment. Also noteworthy is the importance of friction in this task. We noticed that very low friction can lead to slippage and cause the manipulator to overshoot from its desired target. Figure 10 shows the top view of the path the end effector traces in the presence of the obstacle and Figure 11 shows the path without the obstacle. The rigidity of the helicoidal structure does prevent the manipulator from fully conforming to the shape of the obstacle and the tracking error experienced due to that is visible towards the end of the path in Figure 10 and Figure 9 (b). The forces on the manipulator due to the height of the obstacle also hinder the tracking performance.

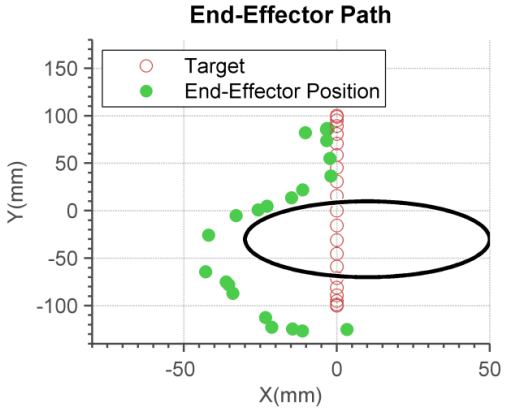


FIG. 10. Path of the end effector in the presence of an obstacle in a line following task.

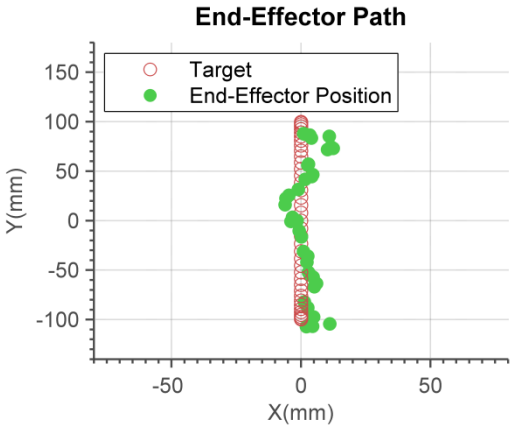


FIG. 11. Path of the end effector in an uninterrupted line following task.

Disturbance rejection during position control

The final test is for showcasing the importance of the end effector feedback for compensating external disturbances. Under external disturbances, the kinematics of the manipulator will get modified as pointed out in the simulations section. Therefore the IK solver needs to provide new solutions for reaching the same target. For this experiment the external forces are applied by hand after the manipulator has converged to the desired target. Due to the compliance of the body, the manipulator can deform without changing the cable lengths. Consequently, the only change in the input to the IK solver is the change in the end effector position. We are not concerned with the exact forces or deformations applied to the manipulator as the experiments are aimed to see if the IK solver can output different solutions just by the change in end effector position and if the solutions can decrease the error in tracking. Clearly there would not be any reaction from an IK solver without the feedback about the end effector position as the actuator configuration remains the same under the disturbance. Besides that, due to the change in kinematics induced by the disturbance, the new inputs provided to the neural network are outliers when compared to the training data. Thus, it is essential to verify that the network can provide decent solutions and more importantly to see that the network does not get saturated. Figure 12 shows the configuration of the manipulator after disturbance and the final configuration derived from the IK solver for 4 different cases. The four trials are conducted in sequence and the solver is updated every 0.5 *sec*. From the figures we can observe that controller is able to respond exceedingly well to the external disturbances. The error in tracking is significantly reduced even with the change in kinematics and from the images we can observe that the manipulator can reach the same target with different configurations. Again, it must be emphasized that the IK solver is never explicitly commanded to reduce the tracking error. The perceived controller is arising purely from the representation and the sample data.

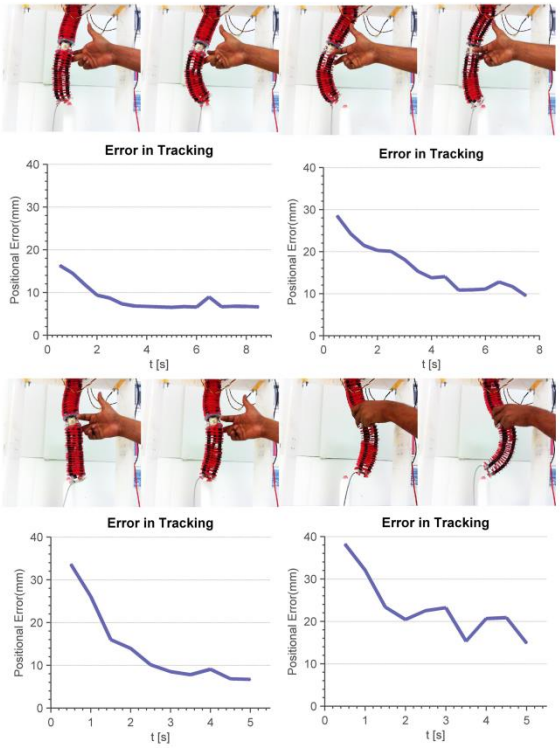


FIG. 12. Disturbance rejection using the kinematic controller. Four cases are shown in the experiment with configuration of the manipulator after disturbance shown first and the final configuration shown in the end. The complete tracking error is shown below.

CONCLUSION AND FUTURE WORK

In conclusion, this paper introduces a machine learning based kinematic controller for continuum robots. Not only is the proposed approach applicable to any form of continuum robot, it has very few parameters to tune and very less sample data requirements compared to existing methods. For instance, for the 6 DoF manipulator, we required just two hours to generate data, learn the network and implement the controller. We consider that it is very important to have controllers with minimal sensory requirements to make its implementation practical and also comparable with rigid robots. In spite of many approximations and sensory handicaps, the proposed IK solver can generate multiple global solutions to the IK problem through an iterative method. Coupled with a unique IK formulation that allows for a feedback mechanism, the kinematic controller can perform highly adaptive motion even

in a completely unstructured environment. The uniqueness in the proposed controller is that in some sense the feedback is used for modifying the kinematic model thereby helping the manipulator adapt in an unstructured environment and this is unlike any formulation possible in rigid robots. To the best of our knowledge this methodology has not been implemented on continuum robots either.

In retrospect, there are few design modifications that we think can improve the performance of the controller. The kinematic controller proposed in this paper is implemented using high gain servo motors. Although this simplifies the low level control of cable lengths, such a system completely nullifies the dynamics of the continuum manipulator. This also makes the control system highly inefficient in terms of energy consumption. One of the potential improvements in the future would be to have low gain controllers in the actuator space, which could significantly improve the performance of the kinematic controller by providing smoother and faster motions. Of course this would require wireless sensors for tracking the end effector position as the cables would affect the manipulator dynamics. Another point of interest is the possibility of using hybrid combination of force and stretch sensors for force and position control as was proposed by Bajo et al.[39]

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